Nuclear Stability

Answer key with explanations

A. Ideas

1. The strong force.

> (a) A very **short-range attraction** between all nucleons (protons and neutrons). (b) Protons repel each other across the whole nucleus, but the strong force acts only between neighbours. Extra **neutrons add attraction without adding repulsion**, so larger nuclei need more of them. (c) Beyond Z = 83, the total proton repulsion is too great for any number of neutrons to hold the nucleus together permanently.

2. N/Z.

> (a) 2/2 = **1.00** (b) 8/8 = **1.00** (c) 30/26 = **1.15** (d) 124/82 = **1.51**. The ratio **rises** with atomic number: heavier stable nuclei need proportionally more neutrons.

B. Predicting decay

3. Decay equations.

> (a) Co-60 has one more neutron than stable Co-59: **β⁻**. $^{60}_{27}\mathrm{Co} \longrightarrow {}^{60}_{28}\mathrm{Ni} + {}^{0}_{-1}\mathrm{e}$
>
> (b) F-18 has one fewer neutron than stable F-19: **β⁺**. $^{18}_{9}\mathrm{F} \longrightarrow {}^{18}_{8}\mathrm{O} + {}^{0}_{+1}\mathrm{e}$
>
> (c) Rn-222 has Z = 86, beyond 83: **α**. $^{222}_{86}\mathrm{Rn} \longrightarrow {}^{218}_{84}\mathrm{Po} + {}^{4}_{2}\mathrm{He}$
>
> (d) P-32 has one more neutron than stable P-31: **β⁻**. $^{32}_{15}\mathrm{P} \longrightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}\mathrm{e}$

4. Doubly magic.

> Both the number of protons **and** the number of neutrons are magic numbers: Ca-40 has 20 and 20; Pb-208 has 82 and 126. Like full electron shells, full nuclear shells give **extra stability**, so these nuclides are especially stable (Pb-208 is the end of the thorium decay series).

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C. Decay series

5. Th-232 to Pb-208.

> α: (232 − 208) ÷ 4 = **6**. Six α lower Z by 12: 90 − 12 = 78. Lead is 82, so Z rises by 4: **4 β⁻**. Check: 90 − 12 + 4 = 82 ✓

6. U-235 to Pb-207.

> α: (235 − 207) ÷ 4 = **7**. Seven α lower Z by 14: 92 − 14 = 78. Lead is 82: **4 β⁻**. Check: 92 − 14 + 4 = 82 ✓

D. Applying the ideas

7. Fluorine-18.

> F-18 has N/Z = 9/9 = 1.00, lower than stable F-19 (10/9 = 1.11): it has **too few neutrons**, so it lies below the band. It converts a proton into a neutron by **emitting a positron**, which then meets an electron in the body and produces the gamma rays a PET scanner detects.

E. Challenge

8. Tritium.

> Tritium has 1 proton and 2 neutrons: N/Z = 2/1 = **2.0**, far above the stable ratios for hydrogen (0 and 1). Too many neutrons: **β⁻ decay**. ${}^{3}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + {}^{0}_{-1}\mathrm{e}$

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